Lecture 2 Consumer Choice Theory I

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Consumer theory: outline

- In the previous lectures we discussed choice in an absolutely abstract setting.
- From this lecture on, we will discuss the choice of *consumers* in a *market economy*.
- Two approaches
  1. Revealed preference approach
  2. Classcial approach (utility maximization and expenditure minimization)
- We will start with revealed preference approach in this lecture and move on to classical approach in the next lecture.

Consumption set

- Assume there are L kinds of commodities.
- Consumers choose commodities from a consumption set

$$X = R_+^L = \{x \in R^L : x_l \geq 0, \forall l = 1, ..., L\}$$

Budget set

- Consumer's are constrained by their wealth (w) so that they can only purchase the commodities they can afford.

Definition

The buget set $B_{p,w} = \{x \in R^L_+ : p \cdot x \leq w\}$ is the feasible consumption bundles for the consumer who faces market prices p and has wealth w.

Demand function

- given price and wealth, a consumer chooses a consumption bundle, which we call *demand*.

Definition

A consumer's **warasian demand correspondence** $x(p, w)$ assigns a chosen consumption bundles for each price-wealth pair (p, w).When $x(p, w)$ is single valued, we call it a **demand function**.

Properties of demand function: homogenous of d0 and Walras' law

Definition

The warasian demand correspondence $x(p, w)$ is **homogeneous of degree zero** if $x(ap, aw) = x(p, w)$ for any p, w and $a > 0$.

Definition

The warasian demand correspondence $x(p, w)$ satisfies the **Walras' law** if for every $p > 0$ and $w > 0$, we have $p \cdot x = w$ for any $x \in x(p, w)$.

Properties of demand function: WARP

Assume that $x(p, w)$ is single-valued, homogeneous of degree zero and satisfies the Walras' law.

Definition

The warasian demand correspondence $x(p, w)$ satisfies the weak **axiom of revealed preference(WARP)** if the following property holds: for any two price-wealth paris (p, w) and (p', w') : if $p \cdot x(p', w') \leq w$, and $x(p', w') \neq x(p, w)$,then $p' \cdot x(p, w) > w'$.

WARP and the law of demand

- The WARP has significant implications for the effects of price changes on demand.
- Imagin a situation in which a change in price is accompanied by a change in consumer's wealth that makes his *initial consumption bundle just affordable at the new price. That is, if consumer is originally facing prices p and wealth w and chooses consumption bundle* $x(p, w)$,then when prices change to p',we adjust consumer's wealth to $w' = p' \cdot x(p, w)$.Thus the wealth adjustment is $\Delta w = (p' - p) \cdot x(p, w)$.
- This kind of wealth adjustment is called **Slutsky wealth compensation.**

WARP and the law of demand

Theorem

suppose demand function $x(p, w)$ is hd0 and satisfies the Walras law. Then $x(p, w)$ satisfies WARP iff: for any compensated price change from an initial situation (p, w) to a new price-wealth pair $(p', w') = (p', p' \cdot x(p, w))$, we have $(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0$, with strict inequality whenver $x(p', w') \neq x(p, w)$.

- The theorem states that WARP and the *compensated law of demand* are equivalent.
- lays the micro fundation why the demand curve is downward sloping.